

# Sales Data Analysis using Power BI

**Prepared By:** Suganya D

**Tool Used:** Microsoft Power BI

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## Project Summary

This project focuses on analyzing sales data using Microsoft Power BI. The project involves importing raw datasets, performing data cleaning and transformation using Power Query, building data models, creating DAX calculations, and designing interactive dashboards. The dashboard provides valuable insights into sales performance, profitability, customer purchasing patterns, and regional trends, enabling data-driven business decisions.

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## Project Objectives

- Analyze overall sales performance.
- Clean and transform raw data into a structured format.
- Build relationships between multiple datasets.
- Create calculated columns and DAX measures.
- Monitor sales trends and profitability.
- Compare actual sales with sales targets.
- Develop an interactive Power BI dashboard.
- Generate business insights for decision-making.

## Dataset Description

The project uses three datasets:

| Dataset        | Description                                                  |
|----------------|--------------------------------------------------------------|
| List of Orders | Contains Order ID, Customer Name, City, State and Order Date |
| Order Details  | Contains Product Category, Quantity, Amount, and Profit      |
| Sales Target   | Contains Monthly Sales Targets by Category                   |

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## Software & Tools Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Microsoft Excel / CSV Files

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## Data Preparation & Transformation

The datasets were imported into Power BI and transformed using Power Query to improve data quality and consistency.

The following transformations were performed:

- Imported all CSV files.
- Retained the first 500 rows for analysis.
- Corrected data types for dates and numeric columns.
- Standardized customer names using Capitalize Each Word.
- Created a Location column by combining City and State.

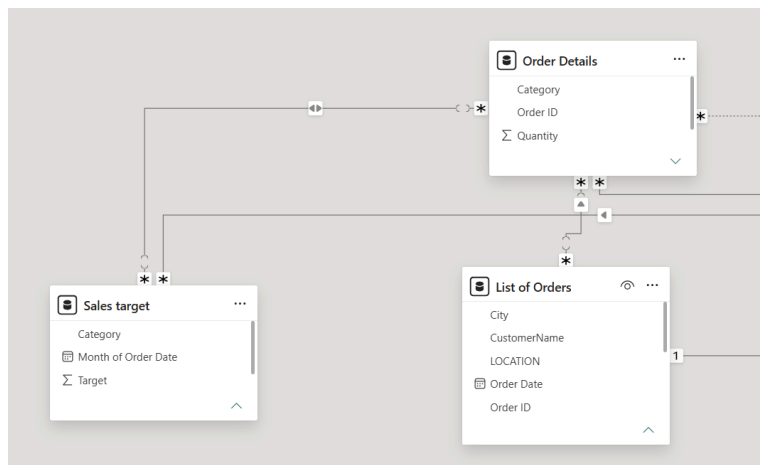
- Calculated Profit Margin.
  - Created a Profit Status column (Profit, Loss, Break-even).
  - Merged List of Orders and Order Details using Order ID.
  - Checked for missing values.
  - Removed duplicate records.
  - Sorted data by Order Date.
  - Applied state-level filtering for analysis.
  - Validated the transformed data before loading into the model.
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## Data Modeling

A one-to-many relationship was created between the **List of Orders** table and the **Order Details** table using the **Order ID** field.

The Sales Target table was connected using Category and Date fields to support target comparison and performance analysis.

The data model enables accurate filtering, aggregation, and interactive reporting.



## DAX Calculations

The following DAX calculations were created:

Calculated Columns

Category Type

Category & " - " & Sub Category

Category Type = 'Order Details'[Category] &"-"& 'Order Details'[Sub-Category]

| Amount | Profit | Quantity | Category    | Sub-Category     | PROFIT MARGIN      | PROFIT STATUS | Category Type                |
|--------|--------|----------|-------------|------------------|--------------------|---------------|------------------------------|
| 65     | 17     | 2        | Clothing    | T-shirt          | 0.261538461538462  | PROFIT        | Clothing-T-shirt             |
| 87     | 4      | 2        | Clothing    | Shirt            | 0.0459770114942529 | PROFIT        | Clothing-Shirt               |
| 50     | 15     | 4        | Clothing    | Leggings         | 0.3                | PROFIT        | Clothing-Leggings            |
| 485    | 29     | 4        | Electronics | Electronic Games | 0.0597938144329897 | PROFIT        | Electronics-Electronic Games |
| 494    | 54     | 4        | Furniture   | Bookcases        | 0.109311740890688  | PROFIT        | Furniture-Bookcases          |
| 42     | 12     | 5        | Clothing    | Hankerchief      | 0.285714285714286  | PROFIT        | Clothing-Hankerchief         |
| 362    | 127    | 1        | Furniture   | Bookcases        | 0.350828729281768  | PROFIT        | Furniture-Bookcases          |
| 353    | 90     | 8        | Clothing    | Saree            | 0.254957507082153  | PROFIT        | Clothing-Saree               |
| 53     | 1      | 4        | Clothing    | Stole            | 0.0188679245283019 | PROFIT        | Clothing-Stole               |
| 26     | 12     | 3        | Clothing    | Hankerchief      | 0.461538461538462  | PROFIT        | Clothing-Hankerchief         |
| 97     | 29     | 2        | Clothing    | Hankerchief      | 0.298969072164948  | PROFIT        | Clothing-Hankerchief         |
| 45     | 13     | 4        | Clothing    | Skirt            | 0.288888888888889  | PROFIT        | Clothing-Skirt               |
| 1560   | 421    | 3        | Clothing    | Trousers         | 0.269871794871795  | PROFIT        | Clothing-Trousers            |
| 133    | 12     | 5        | Clothing    | Stole            | 0.0902255639097744 | PROFIT        | Clothing-Stole               |
| 23     | 2      | 2        | Clothing    | Skirt            | 0.0869565217391304 | PROFIT        | Clothing-Skirt               |
| 637    | 113    | 5        | Clothing    | Saree            | 0.177394034536892  | PROFIT        | Clothing-Saree               |
| 117    | 14     | 3        | Clothing    | Shirt            | 0.11965811965812   | PROFIT        | Clothing-Shirt               |
| 434    | 26     | 11       | Clothing    | Shirt            | 0.0599078341013825 | PROFIT        | Clothing-Shirt               |
| 148    | 72     | 7        | Clothing    | T-shirt          | 0.486486486486486  | PROFIT        | Clothing-T-shirt             |
| 294    | 138    | 2        | Electronics | Printers         | 0.469387755102041  | PROFIT        | Electronics-Printers         |
| 41     | 6      | 5        | Clothing    | Skirt            | 0.146341463414634  | PROFIT        | Clothing-Skirt               |
| 123    | 17     | 3        | Furniture   | Furnishings      | 0.138211382113821  | PROFIT        | Furniture-Furnishings        |
| 523    | 204    | 7        | Clothing    | Trousers         | 0.390057361376673  | PROFIT        | Clothing-Trousers            |
| 433    | 26     | 3        | Electronics | Printers         | 0.0600461893764434 | PROFIT        | Electronics-Printers         |

Revenue per Order

Amount × Quantity

1 Revenue = 'Order Details'[Amount]\*'Order Details'[Quantity]

| Order ID | Amount | Profit | Quantity | Category    | Sub-Category     | PROFIT MARGIN      | PROFIT STATUS | Category Type                | Revenue |
|----------|--------|--------|----------|-------------|------------------|--------------------|---------------|------------------------------|---------|
| B-25604  | 65     | 17     | 2        | Clothing    | T-shirt          | 0.261538461538462  | PROFIT        | Clothing-T-shirt             | ₹ 130   |
| B-25606  | 87     | 4      | 2        | Clothing    | Shirt            | 0.0459770114942529 | PROFIT        | Clothing-Shirt               | ₹ 174   |
| B-25607  | 50     | 15     | 4        | Clothing    | Leggings         | 0.3                | PROFIT        | Clothing-Leggings            | ₹ 200   |
| B-25609  | 485    | 29     | 4        | Electronics | Electronic Games | 0.0597938144329897 | PROFIT        | Electronics-Electronic Games | ₹ 1,940 |
| B-25614  | 494    | 54     | 4        | Furniture   | Bookcases        | 0.109311740890688  | PROFIT        | Furniture-Bookcases          | ₹ 1,976 |
| B-25616  | 42     | 12     | 5        | Clothing    | Hankerchief      | 0.285714285714286  | PROFIT        | Clothing-Hankerchief         | ₹ 210   |
| B-25618  | 362    | 127    | 1        | Furniture   | Bookcases        | 0.350828729281768  | PROFIT        | Furniture-Bookcases          | ₹ 362   |
| B-25619  | 353    | 90     | 8        | Clothing    | Saree            | 0.254957507082153  | PROFIT        | Clothing-Saree               | ₹ 2,824 |
| B-25623  | 53     | 1      | 4        | Clothing    | Stole            | 0.0188679245283019 | PROFIT        | Clothing-Stole               | ₹ 212   |
| B-25624  | 26     | 12     | 3        | Clothing    | Hankerchief      | 0.461538461538462  | PROFIT        | Clothing-Hankerchief         | ₹ 78    |
| B-25625  | 97     | 29     | 2        | Clothing    | Hankerchief      | 0.298969072164948  | PROFIT        | Clothing-Hankerchief         | ₹ 194   |
| B-25628  | 45     | 13     | 4        | Clothing    | Skirt            | 0.288888888888889  | PROFIT        | Clothing-Skirt               | ₹ 180   |
| B-25629  | 1560   | 421    | 3        | Clothing    | Trousers         | 0.269871794871795  | PROFIT        | Clothing-Trousers            | ₹ 4,680 |
| B-25630  | 133    | 12     | 5        | Clothing    | Stole            | 0.0902255639097744 | PROFIT        | Clothing-Stole               | ₹ 665   |
| B-25635  | 23     | 2      | 2        | Clothing    | Skirt            | 0.0869565217391304 | PROFIT        | Clothing-Skirt               | ₹ 46    |
| B-25636  | 637    | 113    | 5        | Clothing    | Saree            | 0.177394034536892  | PROFIT        | Clothing-Saree               | ₹ 3,185 |
| B-25637  | 117    | 14     | 3        | Clothing    | Shirt            | 0.11965811965812   | PROFIT        | Clothing-Shirt               | ₹ 351   |
| B-25642  | 434    | 26     | 11       | Clothing    | Shirt            | 0.0599078341013825 | PROFIT        | Clothing-Shirt               | ₹ 4,774 |
| B-25659  | 148    | 72     | 7        | Clothing    | T-shirt          | 0.486486486486486  | PROFIT        | Clothing-T-shirt             | ₹ 1,036 |
| B-25663  | 294    | 138    | 2        | Electronics | Printers         | 0.469387755102041  | PROFIT        | Electronics-Printers         | ₹ 588   |
| B-25667  | 41     | 6      | 5        | Clothing    | Skirt            | 0.146341463414634  | PROFIT        | Clothing-Skirt               | ₹ 205   |
| B-25668  | 123    | 17     | 3        | Furniture   | Furnishings      | 0.138211382113821  | PROFIT        | Furniture-Furnishings        | ₹ 369   |
| B-25681  | 523    | 204    | 7        | Clothing    | Trousers         | 0.390057361376673  | PROFIT        | Clothing-Trousers            | ₹ 3,661 |
| B-25683  | 433    | 26     | 3        | Electronics | Printers         | 0.0600461893764434 | PROFIT        | Electronics-Printers         | ₹ 1,299 |
| B-25684  | 134    | 42     | 2        | Furniture   | Chairs           | 0.313432835820896  | PROFIT        | Furniture-Chairs             | ₹ 268   |
| B-25685  | 51     | 7      | 2        | Furniture   | Furnishings      | 0.137254901960784  | PROFIT        | Furniture-Furnishings        | ₹ 102   |
| B-25687  | 357    | 139    | 2        | Clothing    | Saree            | 0.389355742296919  | PROFIT        | Clothing-Saree               | ₹ 714   |
| B-25691  | 714    | 56     | 4        | Clothing    | Saree            | 0.0784313725490196 | PROFIT        | Clothing-Saree               | ₹ 2,856 |

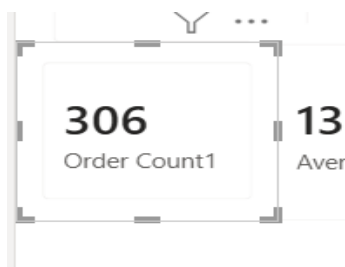
Sales Category

Above Average / Below Average

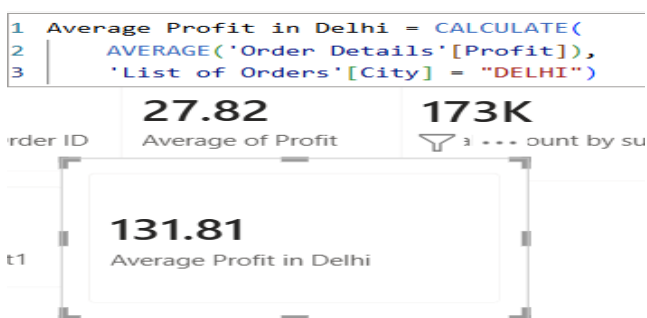
| Structure                         |        | Formatting                                                                                                            |          | Properties  |                  | Sort               | Groups        | Relationships                | Calculations |                |
|-----------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------|--------------------|---------------|------------------------------|--------------|----------------|
| <div><div></div><div></div></div> |        | 1 Sales Category = IF(<br>2 'Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Average","Below Average") |          |             |                  |                    |               |                              |              |                |
| Order ID                          | Amount | Profit                                                                                                                | Quantity | Category    | Sub-Category     | PROFIT MARGIN      | PROFIT STATUS | Category Type                | Revenue      | Sales Category |
| B-25604                           | 65     | 17                                                                                                                    | 2        | Clothing    | T-shirt          | 0.261538461538462  | PROFIT        | Clothing-T-shirt             | ₹ 130        | Below Average  |
| B-25606                           | 87     | 4                                                                                                                     | 2        | Clothing    | Shirt            | 0.0459770114942529 | PROFIT        | Clothing-Shirt               | ₹ 174        | Below Average  |
| B-25607                           | 50     | 15                                                                                                                    | 4        | Clothing    | Leggings         | 0.3                | PROFIT        | Clothing-Leggings            | ₹ 200        | Below Average  |
| B-25609                           | 485    | 29                                                                                                                    | 4        | Electronics | Electronic Games | 0.0597938144329897 | PROFIT        | Electronics-Electronic Games | ₹ 1,940      | Below Average  |
| B-25614                           | 494    | 54                                                                                                                    | 4        | Furniture   | Bookcases        | 0.109311740890688  | PROFIT        | Furniture-Bookcases          | ₹ 1,976      | Below Average  |
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| B-25619                           | 353    | 90                                                                                                                    | 8        | Clothing    | Saree            | 0.254957507082153  | PROFIT        | Clothing-Saree               | ₹ 2,824      | Below Average  |
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| B-25629                           | 1560   | 421                                                                                                                   | 3        | Clothing    | Trousers         | 0.269871794871795  | PROFIT        | Clothing-Trousers            | ₹ 4,680      | Above Average  |
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| B-25637                           | 117    | 14                                                                                                                    | 3        | Clothing    | Shirt            | 0.11965811965812   | PROFIT        | Clothing-Shirt               | ₹ 351        | Below Average  |
| B-25642                           | 434    | 26                                                                                                                    | 11       | Clothing    | Shirt            | 0.0599078341013825 | PROFIT        | Clothing-Shirt               | ₹ 4,774      | Below Average  |
| B-25659                           | 148    | 72                                                                                                                    | 7        | Clothing    | T-shirt          | 0.486486486486486  | PROFIT        | Clothing-T-shirt             | ₹ 1,036      | Below Average  |
| B-25663                           | 294    | 138                                                                                                                   | 2        | Electronics | Printers         | 0.469387755102041  | PROFIT        | Electronics-Printers         | ₹ 588        | Below Average  |
| B-25667                           | 41     | 6                                                                                                                     | 5        | Clothing    | Skirt            | 0.146341463414634  | PROFIT        | Clothing-Skirt               | ₹ 205        | Below Average  |
| B-25668                           | 123    | 17                                                                                                                    | 3        | Furniture   | Furnishings      | 0.138211382113821  | PROFIT        | Furniture-Furnishings        | ₹ 369        | Below Average  |
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| B-25683                           | 433    | 26                                                                                                                    | 3        | Electronics | Printers         | 0.0600461893764434 | PROFIT        | Electronics-Printers         | ₹ 1,299      | Below Average  |
| B-25684                           | 134    | 42                                                                                                                    | 2        | Furniture   | Chairs           | 0.313432835820896  | PROFIT        | Furniture-Chairs             | ₹ 268        | Below Average  |
| B-25685                           | 51     | 7                                                                                                                     | 2        | Furniture   | Furnishings      | 0.137254901960784  | PROFIT        | Furniture-Furnishings        | ₹ 102        | Below Average  |
| B-25687                           | 357    | 139                                                                                                                   | 2        | Clothing    | Saree            | 0.389355742296919  | PROFIT        | Clothing-Saree               | ₹ 714        | Below Average  |

## Measures

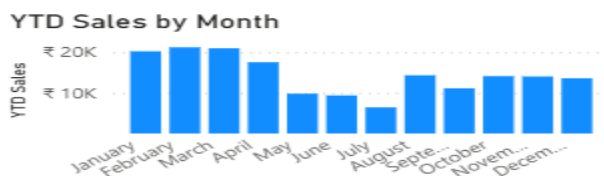
- Order Count



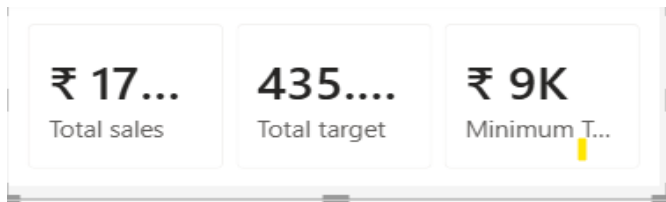
- Average Profit



- Year-to-Date (YTD) Sales



- Total Sales
- Total Profit



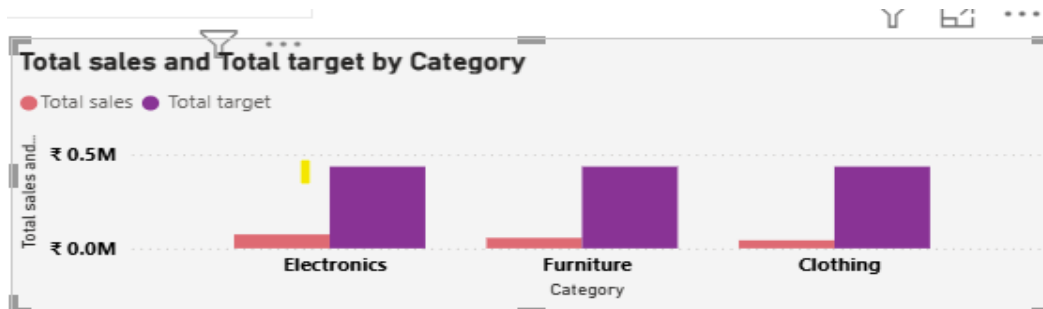
These measures allow dynamic calculations across different dashboard visualizations.

## Dashboard Overview

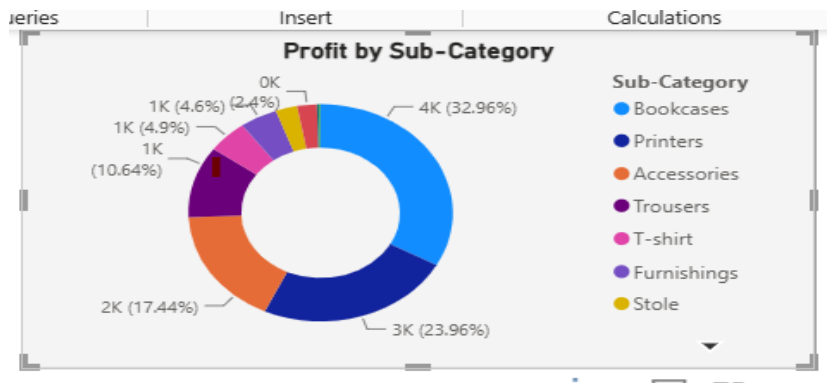
The Power BI dashboard provides an interactive view of business performance using various visualizations.

The dashboard includes:

- Sales vs Target Comparison by Category



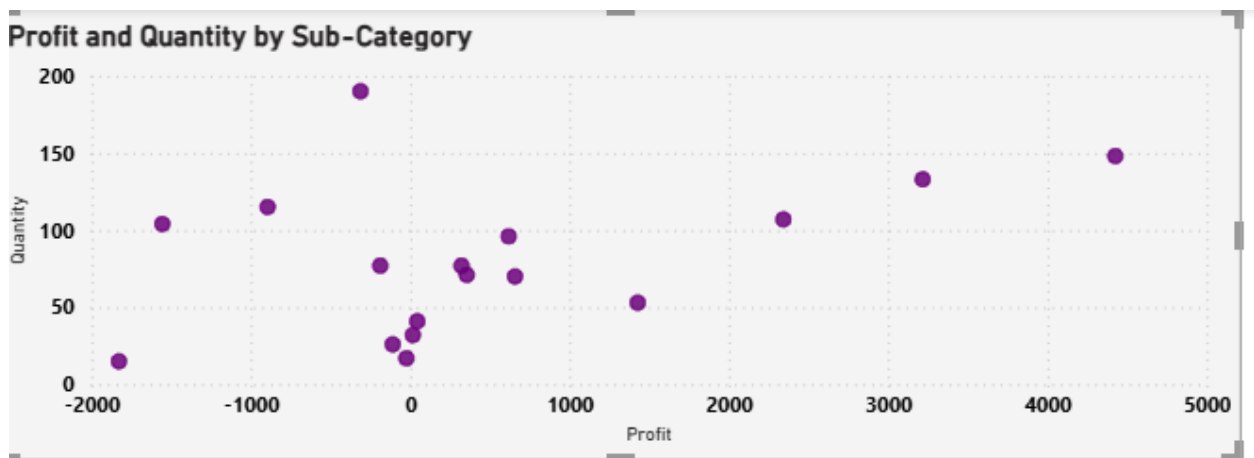
- Max Profit Margin by Sub-Category



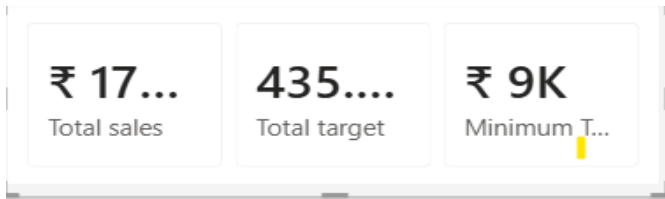
- Monthly Sales Trend



- Profit and Quantity by Sub-Category



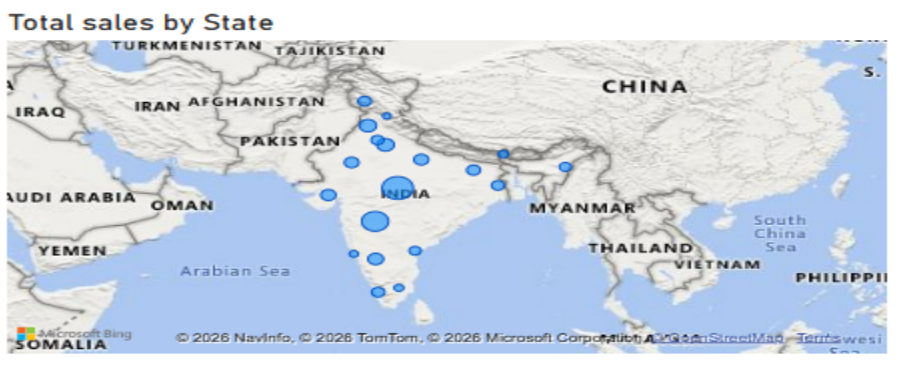
- Comparison of Total Sales Amount and Target:



- Sales Performance Matrix

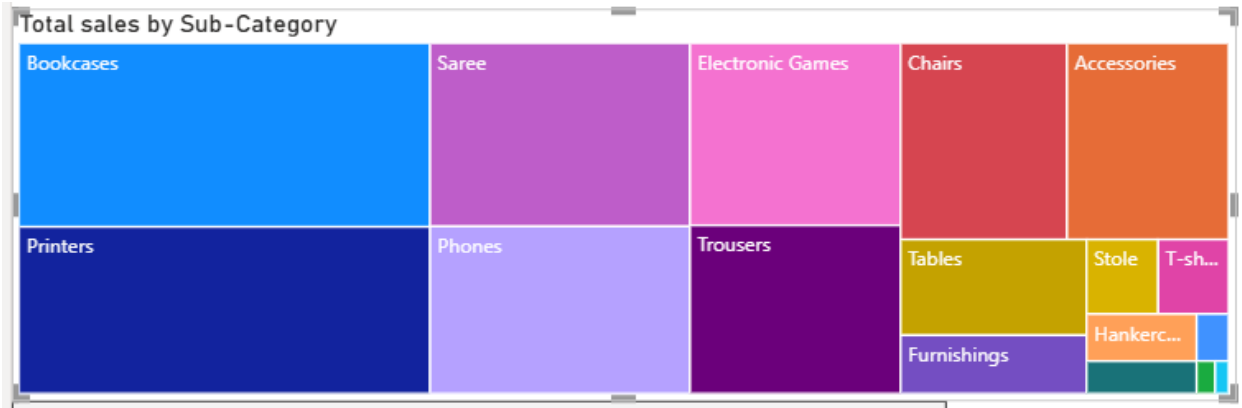
| Category     | Clothing        |                    | Electronics     |                    | Furniture       |                    | Total             |                    |
|--------------|-----------------|--------------------|-----------------|--------------------|-----------------|--------------------|-------------------|--------------------|
| Month        | Total sales     | Total target       | Total sales     | Total target       | Total sales     | Total target       | Total sales       | Total target       |
| January      | ₹ 4,226         | 4,35,900.00        | ₹ 10,533        | 4,35,900.00        | ₹ 5,505         | 4,35,900.00        | ₹ 20,264          | 4,35,900.00        |
| February     | ₹ 1,531         | 4,35,900.00        | ₹ 8,861         | 4,35,900.00        | ₹ 10,863        | 4,35,900.00        | ₹ 21,255          | 4,35,900.00        |
| March        | ₹ 1,996         | 4,35,900.00        | ₹ 9,114         | 4,35,900.00        | ₹ 9,910         | 4,35,900.00        | ₹ 21,020          | 4,35,900.00        |
| April        | ₹ 7,837         | 4,35,900.00        | ₹ 4,191         | 4,35,900.00        | ₹ 5,551         | 4,35,900.00        | ₹ 17,579          | 4,35,900.00        |
| May          | ₹ 2,700         | 4,35,900.00        | ₹ 4,930         | 4,35,900.00        | ₹ 2,198         | 4,35,900.00        | ₹ 9,828           | 4,35,900.00        |
| June         | ₹ 4,011         | 4,35,900.00        | ₹ 4,102         | 4,35,900.00        | ₹ 1,267         | 4,35,900.00        | ₹ 9,380           | 4,35,900.00        |
| July         | ₹ 627           | 4,35,900.00        | ₹ 4,417         | 4,35,900.00        | ₹ 1,399         | 4,35,900.00        | ₹ 6,443           | 4,35,900.00        |
| August       | ₹ 4,785         | 4,35,900.00        | ₹ 3,806         | 4,35,900.00        | ₹ 5,760         | 4,35,900.00        | ₹ 14,351          | 4,35,900.00        |
| September    | ₹ 3,533         | 4,35,900.00        | ₹ 4,254         | 4,35,900.00        | ₹ 3,364         | 4,35,900.00        | ₹ 11,151          | 4,35,900.00        |
| <b>Total</b> | <b>₹ 42,869</b> | <b>4,35,900.00</b> | <b>₹ 74,346</b> | <b>4,35,900.00</b> | <b>₹ 55,886</b> | <b>4,35,900.00</b> | <b>₹ 1,73,101</b> | <b>4,35,900.00</b> |

- Geographic Sales Analysis:

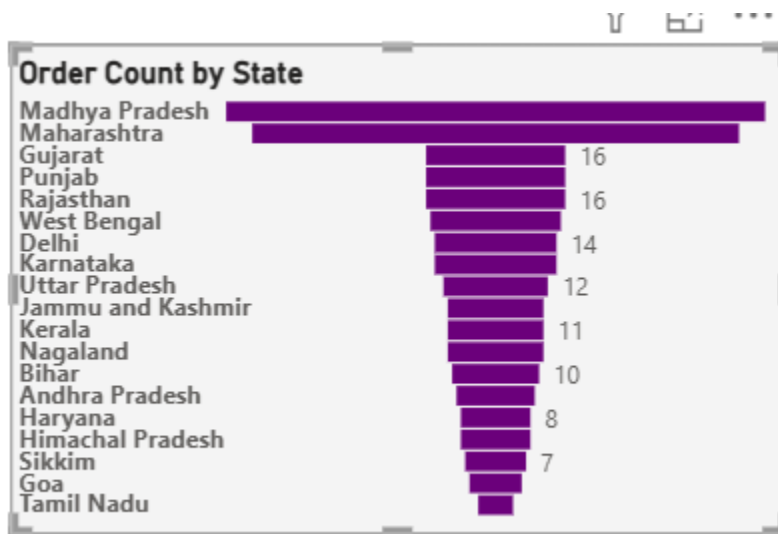


- Sales Distribution by Sub-Category:



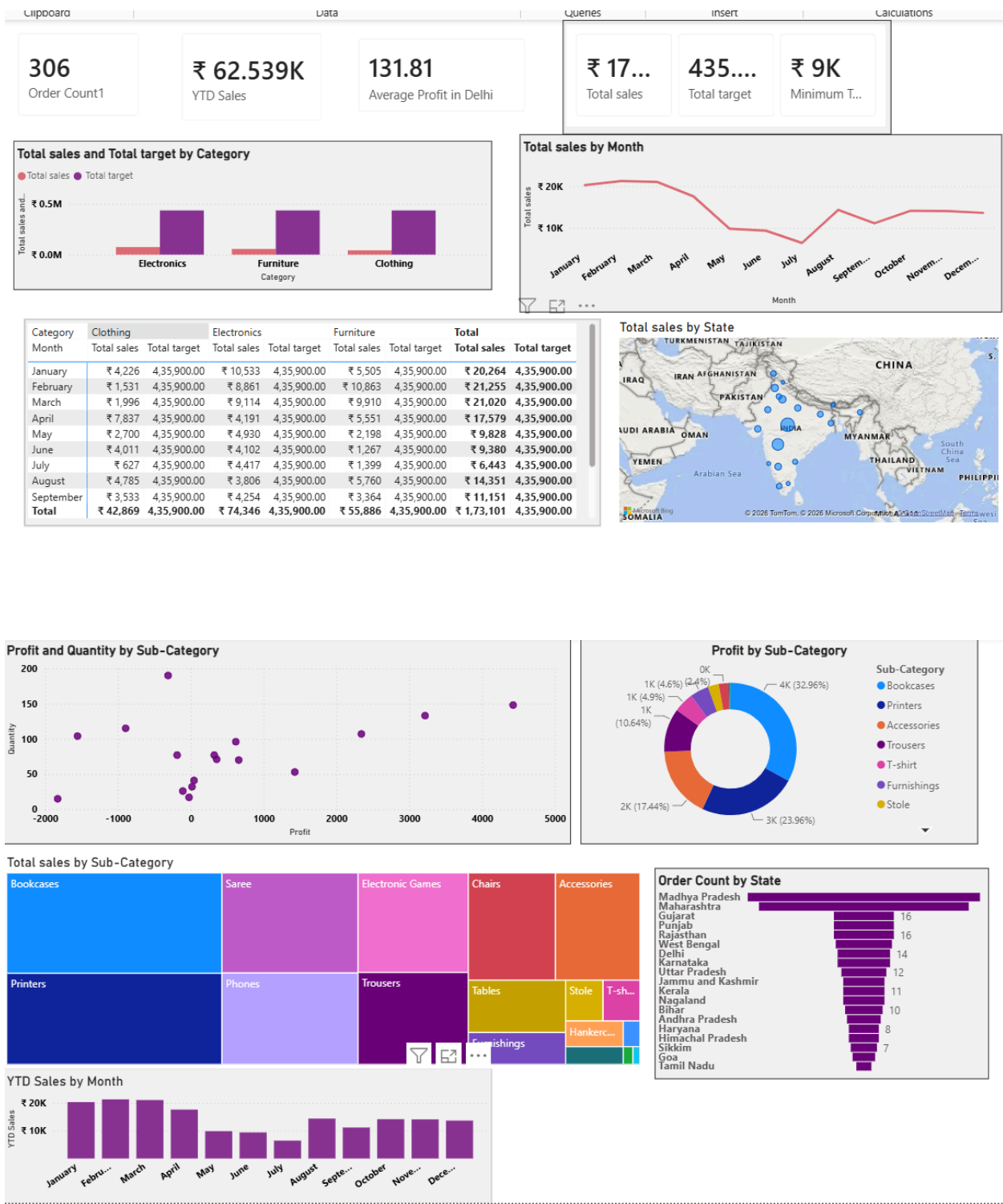


- Order Count Analysis by state:



Users can filter the dashboard by Category, State, Month, and Product to analyze performance from different perspectives.

# Dashboard



Profit and Quantity by Sub-Category

Profit by Sub-Category

Total sales by Sub-Category

Order Count by State

YTD Sales by Month

# **Descriptive Analytics (What Happened?)**

Descriptive analytics summarizes historical business performance using the available sales data.

The dashboard displays total sales, total profit, number of orders, monthly sales trends, and category-wise sales distribution. These visualizations help understand past business performance and identify overall trends.

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# **Diagnostic Analytics (Why Did It Happen?)**

Diagnostic analytics explains the reasons behind business performance.

The dashboard compares profit across product categories and sub-categories, analyzes regional sales performance, and evaluates customer purchasing behavior. These analyses help identify high-performing products and areas requiring improvement.

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# **Predictive Analytics (What Might Happen Next?)**

Predictive analytics uses historical sales patterns to estimate future business performance.

Monthly sales trends and target comparisons provide valuable insights for forecasting future demand, planning inventory, and setting realistic sales targets. Although advanced forecasting models were not implemented, trend analysis supports future business planning.

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# Prescriptive Analytics (What Should Be Done?)

Prescriptive analytics provides recommendations for improving business performance.

Based on the analysis, the organization should:

- Focus on high-profit product categories.
- Improve performance in low-performing regions.
- Optimize inventory based on sales trends.
- Monitor monthly sales targets regularly.
- Develop strategies to increase profitability.
- Strengthen customer engagement through targeted promotions.

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## Business Insights

- Clean and structured data improves reporting accuracy.
- Interactive dashboards simplify business monitoring.
- Sales trends help identify seasonal demand.
- Profit analysis highlights the most valuable product categories.
- Geographic analysis supports regional business expansion.
- DAX measures enable dynamic business reporting.

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## Project Conclusion

This project successfully transformed raw sales data into a structured analytical model using Microsoft Power BI. Data cleaning, transformation, modeling, and DAX calculations enabled the creation of an interactive dashboard that provides meaningful business insights. The project demonstrates how Power BI can support organizations in monitoring performance, identifying trends, and making informed business decisions through data-driven analysis.

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# Thank You

Prepared By: Suganya D